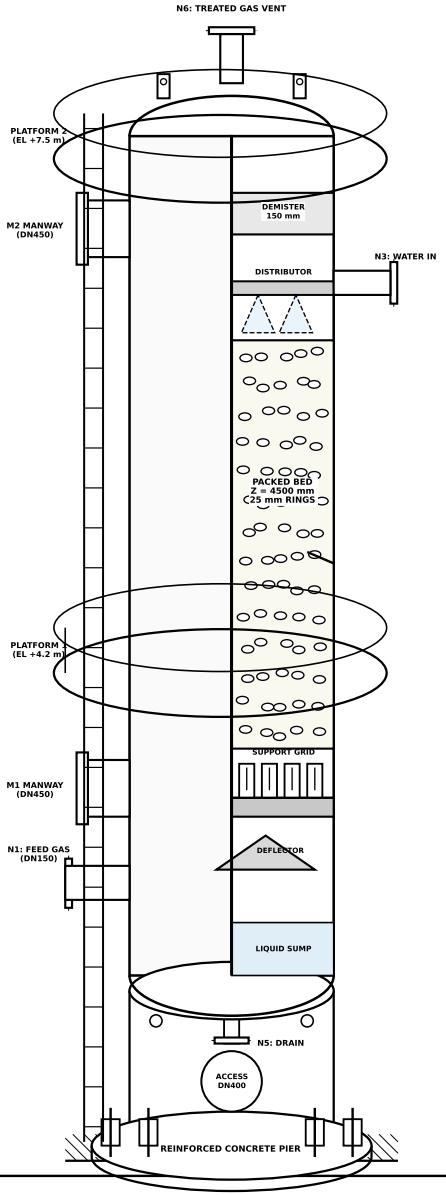
VIEW 1 — 3D ISOMETRIC CUTAWAY GENERAL ARRANGEMENT

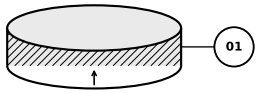
OVERALL MECHANICAL & CUTAWAY INTERNAL ARCHITECTURE — SCALE 1:35



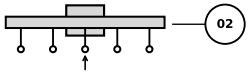
VIEW 2 — 3D EXPLODED INTERNALS STACK

VERTICAL COMPONENT ORDER & ALIGNMENT — SCALE 1:25

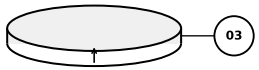
01: KNITTED WIRE MESH DEMISTER (150 mm)



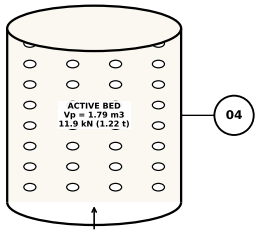
02: HEADER-LATERAL SPRAY DISTRIBUTOR



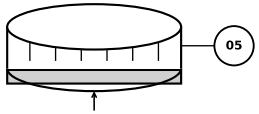
03: BED HOLD-DOWN GRID (ANTI-FLUIDIZATION)



04: 25-mm CERAMIC RASCHIG BED (Z = 4500 mm)



05: GAS-INJECTION SUPPORT GRID (>85% FREE)



06: AERODYNAMIC INLET DEFLECTOR (60 DEG)

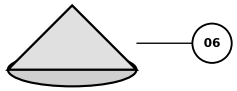


TABLE 1 — 3D INTERNALS EXPLODED BOM

TAG	COMPONENT NAME	MATERIAL / SPEC	QTY
01	Mist Demister Pad	Knitted SS316L / PTFE	1 SET
02	Spray Distributor	SS316L Header-Lateral	1 SET
03	Hold-Down Retaining Grid	SS316L Heavy Mesh	1 SET
04	Ceramic Raschig Bed	Glazed Porcelain 25 mm	1.79 m3
05	Multi-Beam Support Grid	FRP / Ti Gr 2 Chimney	1 SET
06	Inlet Deflector Cone	Titanium Gr 2 (60 DEG)	1 PC

TABLE 2 — EQUIPMENT SPECIFICATION & WEIGHTS

PARAMETER	VALUE	REMARKS / BASIS
Vessel ID & Total Height	**750 mm x 9625 mm**	T/T 7200 mm + Skirt 1800 mm
Active Packed Bed Depth	**4500 mm (1.79 m3)**	25-mm Ceramic Raschig Rings
Dry Column Tare Weight	38.2 kN (3.90 t)	Vessel shell, skirt & internals
Operating Service Weight	42.4 kN (4.32 t)	Dry tare + operating liq holdup
Flooded Upset Weight	68.5 kN (6.98 t)	100% water filled hydrostatic
Foundation Anchorage	8x M24 SA-193 B7	950 mm Bolt Circle Diameter

INTERNAL ASSEMBLY & ERECTION SEQUENCE NOTES

1. **ERECTION: RIG VESSEL VIA TOP LIFTING LUGS; POSITION ON CONCRETE PIER.**
   TORQUE 8x M24 ANCHOR BOLTS TO 450 N*m IN STAR PATTERN. POUR 25 mm GROUT.
2. **SUPPORT GRID: PASS 3 SEGMENTAL PANELS THROUGH M1 MANWAY. SEAT ON**
   50x15 mm LEDGE RING & ISMB 100 BEAM. SECURE WITH PERIMETER J-CLAMPS.
3. **PACKING CHARGING: FILL COLUMN WITH WATER TO DAMPEN IMPACT; DUMP**
   25-mm CERAMIC RASCHIG RINGS RANDOMLY UP TO Z = 4500 mm BED MARK.
4. **DISTRIBUTOR: INSTALL HEADER-LATERAL MATRIX; LEVEL WITHIN +/-1.0 mm.**
5. **DEMISTER: INSTALL KNITTED SS316L/PTFE PAD; TORQUE M1 & M2 MANWAY BLINDS.**

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PACKED-BED ABSORBER (C-101) — 3D ISOMETRIC ASSEMBLY & EXPLODED INTERNALS CAD	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	PICTORIAL 1:25, 1:35 METRIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	PEDD-PBA-06	29 / 09 / 2026

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5